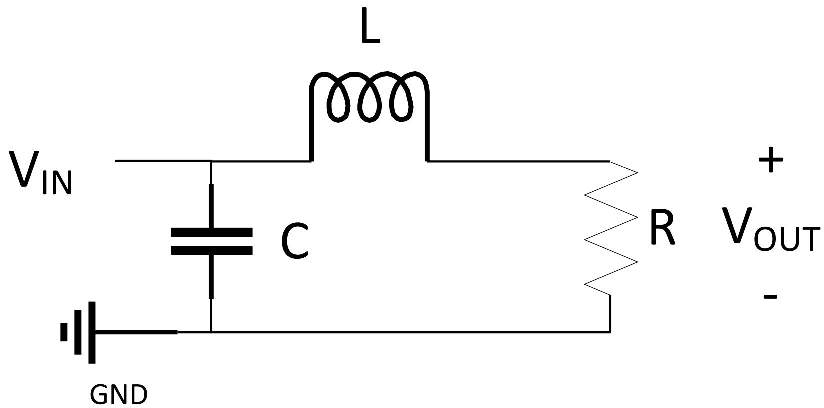
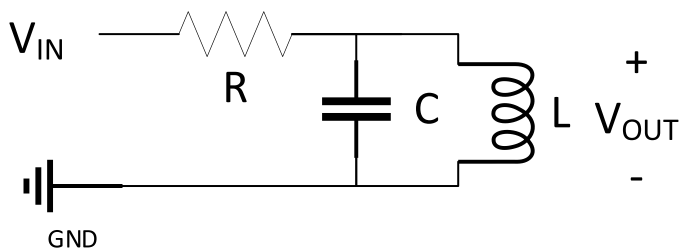
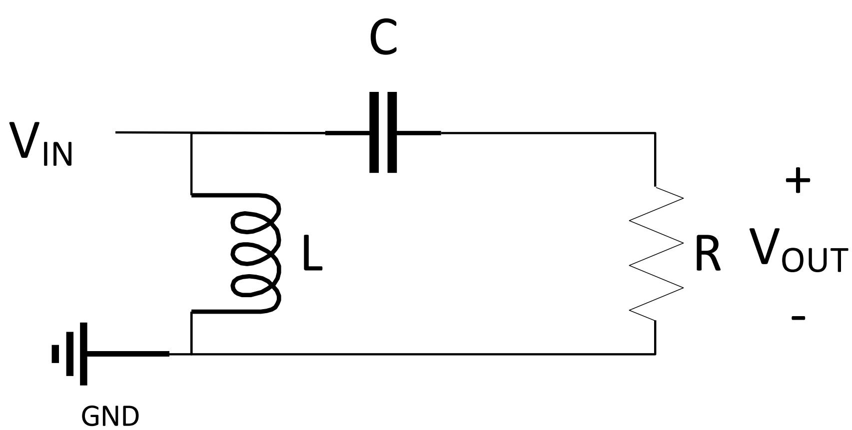
Question 1 – Filter Types

Do you expect the following filters to be high pass, low pass, or band pass? Hint: Redraw the circuit with open or short circuits for inductors and capacitors assuming the input has frequency f=0 or f=∞ Hz.

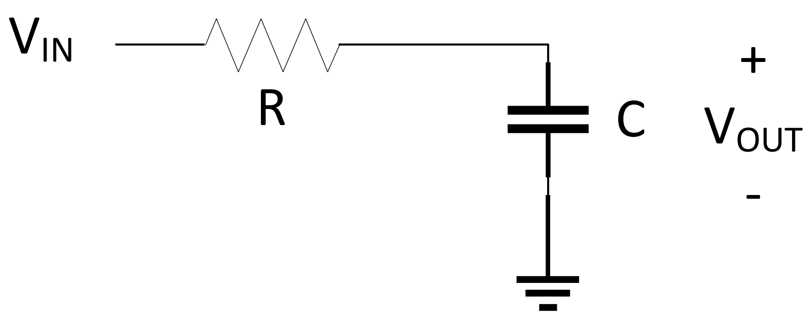






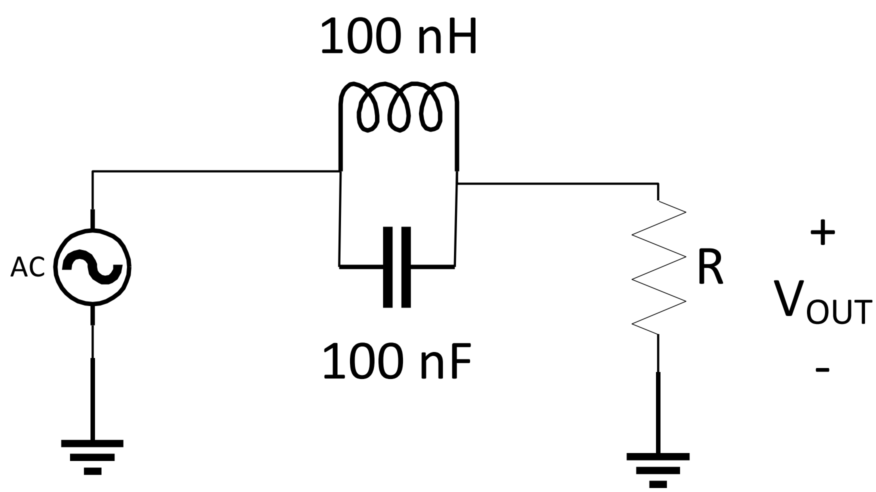
Question 2 – Corner Frequency

We want to design a filter that passes all of the common communication frequencies below 2.5 GHz (radio, TV, cell phone, Wi-Fi) and rejects higher frequencies. If we use a corner frequency of 2.5 GHz and we have a 10 pF capacitor on hand ($10*10^{-12}$ F), choose an appropriate value of R for the low pass filter below. Sketch/generate the Bode Plot of the output power magnitude for this filter as a function of frequency up to 25 GHz.



Question 3 – Notch Filter

What type of filter is pictured below? What frequency does the LC portion of the filter oscillate at (zero or infinite impedance)?



Question 4

If we are receiving a 100 W signal on a line and split it into four lines each with equal power (25 W due to conservation of power/energy), what is the attenuation of the original signal in dB after going through the splitter?